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Turbine support case with axial spokes and retaining members

Abstract

An aircraft engine, has: a turbine; a scroll case having an inlet connected to a source of combustion gases and an outlet connected to the turbine, and a conduit extending from the inlet to the outlet; a bearing housing including a support flange; an exhaust case downstream of the turbine; and a turbine support case secured to the bearing housing and to the exhaust case, the turbine support case having spokes extending along a direction having an axial component, the spokes extending through the scroll case and radially supported by the bearing housing, a spoke having a distal end secured to the support flange via: one or more fasteners, and a retaining member at the distal end, the retaining member defining an abutment face facing an axial direction and circumferentially overlapping the support flange, a portion of the support flange located axially between the distal end and the abutment face.

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Background/Summary

TECHNICAL FIELD

(1) The disclosure relates generally to aircraft engines and, more particularly, to a turbine support case for such engines.

BACKGROUND

(2) In some engine architectures, aerodynamic flow distributors, such as scroll or volute structures, are used to receive combustion gases and to regulate them in a suitable manner before the combustion gases meet stator vanes or rotor blades of the downstream turbine(s). Such structures are subjected to thermal growth, which may have some various effects on surrounding components. Improvements are therefore sought.

SUMMARY

(3) In one aspect, there is provided an aircraft engine, comprising: a turbine including a turbine rotor rotatable about a central axis; a scroll case having an inlet fluidly connected to a source of combustion gases and an outlet fluidly connected to the turbine, and a conduit extending around the central axis from the inlet to the outlet; a bearing housing extending around the central axis, the bearing housing including a support flange; an exhaust case disposed downstream of the turbine; and a turbine support case secured to the bearing housing and to the exhaust case, the turbine support case having spokes distributed around the central axis and extending along a direction having an axial component relative to the central axis, the spokes extending through the scroll case and radially supported by the bearing housing, a spoke of the spokes having a distal end secured to the support flange via: one or more fasteners, and a retaining member at the distal end of the spoke, the retaining member defining an abutment face facing an axial direction relative to the central axis and circumferentially overlapping the support flange, a portion of the support flange located axially between the distal end of the spoke and the abutment face.

(4) The aircraft engine described above may include any of the following features, in any combinations.

- (5) In some embodiments, the retaining member is a hook at the distal end of the spoke, the hook having a first hook section protruding axially from the distal end and a second hook section protruding transversally from the first hook section, the abutment face defined by the second hook section.
- (6) In some embodiments, the hook and the spoke are parts of a single monolithic body of the spoke.
- (7) In some embodiments, the support flange defines a member-receiving aperture sized to receive the retaining member.
- (8) In some embodiments, the retaining member defines a slot for receiving the portion of the support flange, the slot facing a circumferential direction opposite to a direction of rotation of the turbine rotor.
- (9) In some embodiments, the scroll case includes vanes extending in a direction having an axial component relative to the central axis and across the conduit.
- (10) In some embodiments, each of the spokes extends within a respective one of the vanes.
- (11) In some embodiments, the spokes are free of connection to the vanes.
- (12) In some embodiments, the turbine support case includes a wall extending around the central axis, the spokes protruding from the wall.
- (13) In some embodiments, the wall axially overlaps at least a portion of the turbine, the turbine support case having a rear flange secured to a flange of the exhaust case.
- (14) In another aspect, there is provided a turbine assembly, comprising: a turbine including a turbine rotor rotatable about a central axis; a support structure; a scroll case for receiving combustion gases and for directing the combustion gases to the turbine, the scroll case having a conduit extending around the central axis; and a turbine support case having spokes distributed around the central axis, the spokes extending through the conduit of the scroll case and radially supported by the support structure via: one or more fasteners, and an axial locking engagement defined between distal ends of the spokes and a portion of the support structure.
- (15) The turbine assembly described above may include any of the following features, in any combinations.
- (16) In some embodiments, the axial locking engagement is defined by hooks provided at distal ends of the spokes.
- (17) In some embodiments, the hooks have first hook sections protruding axially from the distal ends and second hook sections protruding transversally from the first hook sections, the second hook sections axially overlapping the portion of the support structure.
- (18) In some embodiments, the hooks and the spokes are parts of a single monolithic body of the turbine support case.
- (19) In some embodiments, the portion of the support structure is an annular flange defining apertures, the hooks received through the apertures.
- (20) In some embodiments, the hooks define slots for receiving a portion of the annular flange.
- (21) In some embodiments, the slots are facing a circumferential direction opposite to a direction of rotation of the turbine rotor.
- (22) In some embodiments, the scroll case includes vanes extending in a direction having an axial component relative to the central axis and across the conduit.
- (23) In some embodiments, each of the spokes extends within a respective one of the vanes.
- (24) In some embodiments, the spokes are free of connection to the vanes.

Description

DESCRIPTION OF THE DRAWINGS

(1) Reference is now made to the accompanying figures in which:

- (2) FIG. 1 is a schematic side view of an aircraft engine;
- (3) FIG. 2 is a side cross-sectional view of a portion of the aircraft engine of FIG. 1 illustrating a hot section of the aircraft engine;
- (4) FIG. 3 is an enlarged view of a portion of FIG. 2;
- (5) FIG. 4 is a three dimensional exploded view of a turbine assembly for the aircraft engine of FIG. 1, including a bearing housing, a scroll case, and a turbine support case;
- (6) FIG. 5 is a three dimensional view of the turbine support case of FIG. 4;
- (7) FIG. 6 is a cross-sectional view taken on a plane normal to a central axis of the aircraft engine of FIG. 1, illustrating the turbine support case and the scroll case;
- (8) FIG. 7 is an enlarged view of a portion of FIG. 3;
- (9) FIG. 8 is an enlarged view of a portion of FIG. 7;
- (10) FIG. 9 is a three-dimensional view illustrating a retaining member of the turbine support case of FIG. 3; and
- (11) FIGS. 10 to 12 are front views illustrating an assembly sequence of the turbine support case.

DETAILED DESCRIPTION

(12) Referring to FIG. 1, an aircraft engine **10** is schematically shown. The aircraft engine **10** comprises a thermal engine module **11** including one or more internal combustion engine(s), drivingly engaged to a rotatable load **12**, herein depicted as a propeller, via an output shaft **13**. It will be appreciated that the thermal engine module **11** may include any suitable engine, such as a gas turbine engine, a rotary engine, a piston engine, and so on. The output shaft **13** may correspond to an engine shaft of the thermal engine module **11**. The thermal engine module **11** may include any engine having at least one combustion chamber of varying volume. For instance, the thermal engine module **11** may comprise one or more piston engine(s) or one or more rotary engine(s) (e.g., Wankel engines). The aircraft engine **10** further includes a compressor **14** having a compressor inlet receiving ambient air from the environment E outside the aircraft engine **10** and a compressor outlet fluidly connected to an air inlet of the thermal engine module **11**. The compressor **14** outputs compressed air from the compressor outlet to the thermal engine module **11** via a compressed air conduit **16** and a manifold **17**. The compressed air conduit **16** and the manifold **17** may include any suitable arrangement of pipes configured to distribute compressed air between the different combustion chambers of the thermal engine module **11**. Any other suitable configurations used to supply compressed air to the thermal engine module **11** are contemplated without departing from the scope of the present disclosure. The aircraft engine **10** further includes a turbine assembly **15** having an axially facing turbine inlet **15A** fluidly connected to an engine outlet of the thermal engine module **11**. The turbine **15** has a turbine exhaust case **15B** via which combustion gases are expelled to the environment E. The turbine exhaust case **15B** may include a tailpipe or any other suitable structures (e.g., exhaust mixer) for discharging the combustion gases from the aircraft engine **10**. In some embodiments, the engine **10** may be a hybrid engine including an electric motor drivingly engaged to the output shaft **13** to assist the thermal engine module **11** in driving the output shaft **13** and the rotatable load (e.g., propeller **12**) mounted thereto.

(13) Referring jointly to FIGS. 1-2, in one or more embodiment(s), the turbine **15** includes an axial turbine having successive rows of rotor(s) **15C** and stator(s) **15D** disposed in alternation along a central axis A of the aircraft engine **10**. The rotor(s) **15C** may include rotor blades mounted to rotor discs. The stator(s) **15D** may include stator vanes secured at opposite ends to inner and outer shrouds. In other words, the turbine **15** may include a plurality of stages each including a stator and a rotor. The rotors **15C** of the turbine **15** are in driving engagement with a turbine shaft **15E**. The turbine shaft **15E** may be drivingly engaged to the output shaft **13**, which may correspond to the engine shaft of the thermal engine module **11**. Therefore, the turbine **15** may compound power with the thermal engine module **11** to drive the rotatable load **12**. In other words, the turbine shaft **15E** may be drivingly engaged to the engine shaft of the thermal engine module **11** via suitable gearing. In the embodiment shown, the turbine shaft **15E** is drivingly engaged to a compressor shaft of the

compressor **14**. Thus, the turbine **15** may drive both the rotatable load **12** and the compressor **14**. In the exemplified embodiment, the engine shaft of the thermal engine module **11**, the output shaft **13**, and the turbine shaft **15E** are all coaxial about the central axis A. However, in other configurations, the turbine **15** and/or the compressor **14** may have respective shafts radially offset from one another relative to the central axis A.

(14) As shown in FIG. 1, the engine outlet of the thermal engine module **11** is fluidly connected to an exhaust manifold **18** that receives combustion gases outputted by the combustion chambers or by a combustor of the thermal engine module **11**. The exhaust manifold **18** collects the combustion gases from the different combustion chambers and flows these combustion gases to a combustion engine exhaust pipe **19** that feeds the combustion gases to the turbine **15**. In other words, the engine outlet of the thermal engine module **11** is fluidly connected to the turbine inlet **15A** via the exhaust manifold **18** and the combustion engine exhaust pipe **19**. Any other suitable configurations used to supply combustion gases to the turbine **15** are contemplated without departing from the scope of the present disclosure.

(15) As schematically depicted by the flow arrows in FIG. 1, the combustion gases are flowing within the combustion engine exhaust pipe **19** and reach the turbine **15** in a direction being mainly radial relative to the central axis A and which may include a circumferential component relative to the central axis A. However, the turbine **15** includes an axial turbine and therefore the turbine inlet **15A** receives the combustion gases along a direction being mainly axial relative to the central axis A. To redirect the combustion gases from a direction being mainly radial to a direction being mainly axial, that is, to decrease a radial component of a direction of the combustion gases, the aircraft engine **10** further includes a scroll case **20** that regulates and reorients the combustion gases so that they meet an upstream most of the stages of the turbine **15** at the most appropriate angle of attack. In the embodiment shown, the flow of combustion gases exiting the scroll case **20** meets a first stage rotor **15C** of the turbine **15** before meeting a stator thereof. The scroll case **20** may therefore be used to adequately orient the combustion gases at the most appropriate angle to meet the upstream-most airfoils of the turbine **15**, which are herein part of one of the first stage rotors **15C**.

(16) Referring to FIG. 3, as shown in the exemplified embodiment, the scroll case **20** may be provided in form of a unitary body or mono-case comprising a conduit **21** extending around the central axis A from an inlet **22** to an outlet **23**. The inlet **22** is fluidly connected to the combustion engine exhaust pipe **19**, whereas the outlet **23** is fluidly connected to the turbine inlet **15A** (FIG. 2) of the turbine **15**. According to the illustrated embodiment, the inlet **22** of the conduit **21** has a tangential component and the outlet **23** is an annular outlet facing axially in a rearward direction and in alignment with an annular gas path **15F** of the turbine **15**. This configuration allows injecting the combustion gases in a direction being mainly axial relative to the central axis A to meet the axial inlet of the turbine **15**. Vanes **24** may be provided in the conduit **21** to direct and regulate the flow of combustion gases. The vanes **24** may be omitted in some embodiments. The conduit **21** of the scroll case **20** is in this embodiment disposed axially forwardly of the turbine **15**.

(17) The conduit **21** comprises a non-axisymmetric portion extending downstream from the inlet **22** and spiraling towards the central axis A. As it progresses circumferentially around the central axis A, the non-axisymmetric portion of the conduit **21** transitions or merges with an axisymmetric portion, which forms a 360 degrees axisymmetric structure around the central axis A. The axisymmetric portion extends downstream from the non-axisymmetric portion to the outlet **23**.

(18) The inventors have found that in engine running conditions, the thermal distortions are non-uniform in the non-axisymmetric portion of the scroll case **20**. Consequently, using the scroll case **20** to secure the turbine exhaust case **15B** may increase tip clearance of the rotors **15C** of the turbine **15**. In other words, radial thermal growth of the scroll case **20** during use of the engine may move the turbine exhaust case **15B** radially outwardly, thus pulling radially on shrouds disposed around the rotors **15C**. This may increase tip clearance and, as a result, may impair performance.

As will be seen hereafter, a turbine support case arrangement may be used to alleviate these drawbacks.

(19) As illustrated on FIG. 3, a compressor case **14A** of the compressor **14** is radially supported by a bearing housing **30**. It will be appreciated that that any suitable support structure may be used for support the compressor case **14A**. For instance, the support structure may be any static component of the engine, such as a support flange and so on. Bearings **31** are rollingly engaged to the bearing housing **30** and radially support a shaft of the engine. The scroll case **20** is secured to a rear end **32** of the bearing housing **30**. In the exemplified embodiment, the scroll case **20** has a radially-inner wall **25** that defines a flange at its rear end. The flange of the radially-inner wall **25** is received within an annular groove defined by the rear end **32** of the bearing housing **30**. Other configurations are however contemplated. Therefore, the scroll case **20** may not rely on the turbine exhaust case **15B** for structural support.

(20) In the disclosed embodiment, a turbine support case **40** is used to secure the turbine exhaust case **15B** to the compressor case **14A** of the compressor **14**. As will be explained below, the turbine support case **40** is independent from the scroll case **20** such that thermal growth of the scroll case **20** may not be transmitted to the turbine exhaust case **15B**. Therefore, the turbine exhaust case **15B** is secured to the compressor case **14A** via the turbine support case **40** independently of the scroll case **20**. In the present disclosure, the expression “independent” or “independently” in “independently of the scroll case **20**” implies that a load path extends from the compressor case **14A** to the turbine exhaust case **15B** through the turbine support case **40** without intersecting the scroll case **20**. The scroll case **20** is therefore free from intersection to the load path from the compressor case **14A** to the turbine exhaust case **15B**. The scroll case **20** is thus not part of the load path from the compressor case **14A** to the turbine exhaust case **15B** and loads generated by the turbine **15** on the turbine exhaust case **15B** are transmitted to the compressor case **14B** via the turbine support case **40** without assistance from the scroll case **20**. The scroll case **20** is thus outside the load path that extends through the turbine support case **40**. The scroll case **20** may thus be structurally floating relative to the turbine support case **40**.

(21) Referring to FIG. 4, the turbine support case **40** has a portion that axially overlaps the scroll case **20** and is secured to an annular member **41**, which is itself secured to the bearing housing **30** or any other suitable support structure. More specifically, the annular member **41** has a flange **42** secured (e.g., bolted) to a first flange **33** of the bearing housing **30**. The bearing housing **30** further has a second flange **34**, which may be disposed radially outwardly of the first flange **33** and axially offset from the first flange **33**, for being secured (e.g., bolted) to a mating flange of the compressor case **14A**.

(22) The turbine support case **40** includes a wall **43** extending around the central axis A. The wall **43** may be cylindrical, frustoconical, or any other suitable shape. The wall **43** may extend a full circumference around the central axis A. The turbine support case **40** further includes spokes **44** protruding from the wall **43**. More specifically, the turbine support case **40** includes an annular axial wall **45** extending radially inwardly from the wall **43**. The spokes **44** protrude in a direction having an axial component relative to the central axis A from the annular axial wall **45** and away from the wall **43**. The spokes **44** may be parallel to the central axis A. An annular flange **46** is provided at a rear end of the wall **43** and is secured (e.g., bolted) to a mating flange **15G** (FIG. 3) of the turbine exhaust case **15B**.

(23) As shown in FIG. 3, the wall **43** axially overlaps at least a portion of the turbine **15**. A containment ring **50** may be secured to the flange **15G** of the turbine exhaust case **15B** via containment ring flange **51**, which may be sandwiched between the annular flange **46** of the turbine support case **40** and the flange **15G** of the turbine exhaust case **15B**. The containment ring **50** is, in this embodiment, disposed radially between the wall **43** of the turbine support case **40** and at least one of the rotors **15C** of the turbine **15**.

(24) The spokes **44**, six in the illustrated embodiment, but more or less may be used, extend from

proximal ends **44A** at the annular axial wall **45** to distal ends **44B**. The distal ends **44B** of the spokes **44** are secured to the annular member **41** as will be explained further below. The distal ends **44B** of the spokes define threaded apertures **44C** (FIG. 5) threadingly engageable by fasteners **47** (e.g., bolts) extending through correspondingly-shaped apertures defined through the annular member **41** and threadingly engaged to the threaded apertures **44C** for securing the spokes **44** to the annular member **41**, which is itself secured to the bearing housing **30**.

(25) Referring to FIGS. 4 and 6, in the embodiment shown, each of the spokes **44** is received within a respective one of the hollow vanes **24** of the scroll case **20**. The spokes **44** therefore axially overlap the vanes **24**. Thus, the spokes **44** may be isolated from combustion gases flowing through the scroll case **20** by the vanes **24**. The spokes **44** may be free of connection to the vanes **24**. In other words, outer surfaces of the spokes **44** may be free of contact with inner surfaces of the vanes **24**. An annular gap may be provided between the inner surface of each vanes **24** and the associated spokes **44** extending internally therethrough. The vanes **24** may move axially, radially, and/or circumferentially relative to the spokes **44** without transferring any forces to the spokes **44**, and vice versa. Put differently, the scroll case **20** is free from direct connection to the turbine support case **40**. In other words, the scroll case **20** is free of contact, attachment, so on with the turbine support case **40**. The spokes **44** of this embodiment have an elongated, airfoil-like shape to substantially match a shape of the vanes **24**. However, the shape of the spokes **44** may be different. The spokes **44** may be circular, oval, square, rectangular in cross-section and so on, without departing from the scope of the present disclosure.

(26) In some conditions, a torsional load may be applied to the turbine support case **40**. In such a situation, it is desired to prevent this load from shearing the fasteners **47** since this may impede the integrity of the connection between the turbine support case **40** and the associated supporting structure (e.g., the bearing housing **30**). The turbine support case **40** of the present disclosure may at least partially alleviate these drawbacks.

(27) Referring more particularly to FIG. 5, in the embodiment shown, the spokes **44** are secured to a portion of the bearing housing **30**, herein to the annular member **41** that is itself secured to the bearing housing **30** via the mating flanges (i.e., first flange **33** and flange **42**), via the fasteners **47** and via retaining members **48** located at the distal ends **44B** of the spoke **44**. It will be appreciated that not all of the spokes **44** may present such a retaining member. In some embodiments, only one of the spokes **44** may present a retaining member. Herein, a retaining member **48** is provided to all of the spokes **44**.

(28) As depicted in FIGS. 7-9, a portion of the annular member **41** is located axially between the distal ends **44B** of the spokes and the retaining members **48**. The retaining member **48** defines an abutment face **48A** (FIG. 9) facing an axial direction relative to the central axis A and circumferentially overlapping the annular member **41**. As shown in FIG. 5, the retaining members **48** may be configured as hooks **49** including first hook sections **49A** protruding axially from the distal ends **44B** of the spokes **44** and second hook sections **49B** protruding transversally from the first hook sections **49A**. The abutment faces **48A** are defined by the second hook sections **49B**. The first and second hook sections **49A**, **49B** jointly define an L-shape. In this embodiment, the hooks **49** and the spokes **44** are parts of a single monolithic body of the spokes **44**, and of the turbine support case **40**. However, this need not be the case and the hooks may be secured at the distal ends **44B** of the spokes **44** via any suitable connections, such as, for instance, locking engagement, fasteners, and so on.

(29) Referring more particularly to FIGS. 10-12, the annular member **41** defines apertures **41A** sized to receive the retaining members **48** (e.g., hooks **49**). The retaining members **48** may define slots **49C** (FIG. 5) located axially between the second hook sections **49B** and the distal ends **44B** of the spokes **44**. The slots **49C** are sized to receive the annular member **41** therein. Stated differently, the slots **49C** may have a size sufficient to accommodate a thickness of the annular member **41**. The slots **49C** may face a circumferential direction such that the rotation of the rotor **15C** pushes

the annular member **41** in engagement within the slots **49C**. This direction may be the same as a direction of rotation of the rotor **15C**.

(30) Stated differently, the turbine support case **40** defines an axial locking engagement defined between the distal ends **44B** of the spokes **44** and the support structure, which corresponds herein to the annular member **41** secured to the bearing housing **30**. The axial locking engagement is provided by the hooks at the distal ends **44B** of the spokes **44**. These hooks are used to axially lock the turbine support case **40** to the annular member **41** like a bayonet style locking engagement.

(31) As shown in FIGS. **10-12**, to axially lock the turbine support case **40** to the annular member **41**, which is itself secured to the bearing housing **30**, the turbine support case **40** is moved axially relative to the annular member **41** until the hooks **49** are received through the apertures **41A** defined through the annular member **41** (see FIG. **10**). At which point, and as illustrated in FIG. **11**, the turbine support case **40** is rotated about direction **D1** relative to the central axis **A** until the annular member **41** is engaged within the slots **49C** defined by the hooks **49**. Then, as shown in FIG. **12**, the distal ends **44B** of the turbine support case **40** may be secured to the annular member **41** via the fasteners **47**.

(32) The retaining members **48**/hooks **49** may prevent the turbine support case **40**, which supports the hot section of the aircraft engine **10** from separating, thus mitigating the risk of not containing the internal components. The hooks **49** are designed to be assembled like a bayonet structure. When the fasteners **47** used to assemble the axial spoke are secured in place the hooks **49** remain engaged. The hooks **49** are oriented such as the sense of rotation of the engine ensures the hooks **49** are always engaging in the mating slot/apertures **41A**. The size of the hooks **49** is designed to meet conditions of unanticipated ultimate load. The exemplified bayonet structure provides a secondary attachment structure in addition to the primary bolting structure and is thus helpful in maintaining the integrity of the connection between the turbine support case **40** and the bearing housing **30**. Notably, it contributes to prevent the transmission of shear forces to the bolts **47**. It can thus be used as a means to protect the integrity of the bolts **47** while providing axial retention redundancy.

(33) It is noted that various connections are set forth between elements in the preceding description and in the drawings. It is noted that these connections are general and, unless specified otherwise, may be direct or indirect and that this specification is not intended to be limiting in this respect. A coupling between two or more entities may refer to a direct connection or an indirect connection. An indirect connection may incorporate one or more intervening entities. The term “connected” or “coupled to” may therefore include both direct coupling (in which two elements that are coupled to each other contact each other) and indirect coupling (in which at least one additional element is located between the two elements).

(34) It is further noted that various method or process steps for embodiments of the present disclosure are described in the preceding description and drawings. The description may present the method and/or process steps as a particular sequence. However, to the extent that the method or process does not rely on the particular order of steps set forth herein, the method or process should not be limited to the particular sequence of steps described. As one of ordinary skill in the art would appreciate, other sequences of steps may be possible. Therefore, the particular order of the steps set forth in the description should not be construed as a limitation.

(35) Furthermore, no element, component, or method step in the present disclosure is intended to be dedicated to the public regardless of whether the element, component, or method step is explicitly recited in the claims. As used herein, the terms “comprises”, “comprising”, or any other variation thereof, are intended to cover a non-exclusive inclusion, such that a process, method, article, or apparatus that comprises a list of elements does not include only those elements but may include other elements not expressly listed or inherent to such process, method, article, or apparatus.

(36) While various aspects of the present disclosure have been disclosed, it will be apparent to those of ordinary skill in the art that many more embodiments and implementations are possible

within the scope of the present disclosure. For example, the present disclosure as described herein includes several aspects and embodiments that include particular features. Although these particular features may be described individually, it is within the scope of the present disclosure that some or all of these features may be combined with any one of the aspects and remain within the scope of the present disclosure. References to “various embodiments,” “one embodiment,” “an embodiment,” “an example embodiment,” etc., indicate that the embodiment described may include a particular feature, structure, or characteristic, but every embodiment may not necessarily include the particular feature, structure, or characteristic. Moreover, such phrases are not necessarily referring to the same embodiment. The use of the indefinite article “a” as used herein with reference to a particular element is intended to encompass “one or more” such elements, and similarly the use of the definite article “the” in reference to a particular element is not intended to exclude the possibility that multiple of such elements may be present.

(37) The embodiments described in this document provide non-limiting examples of possible implementations of the present technology. Upon review of the present disclosure, a person of ordinary skill in the art will recognize that changes may be made to the embodiments described herein without departing from the scope of the present technology. Yet further modifications could be implemented by a person of ordinary skill in the art in view of the present disclosure, which modifications would be within the scope of the present technology.

Claims

1. An aircraft engine, comprising: a turbine including a turbine rotor rotatable about a central axis; a scroll case having an inlet fluidly connected to a source of combustion gases and an outlet fluidly connected to the turbine, and a conduit extending around the central axis from the inlet to the outlet; a bearing housing extending around the central axis, the bearing housing including a support flange; an exhaust case disposed downstream of the turbine; and a turbine support case secured to the bearing housing and to the exhaust case, the turbine support case having spokes distributed around the central axis and extending along a direction having an axial component relative to the central axis, each of the spokes extending through the scroll case and radially supported by the bearing housing, a spoke of the spokes having a distal end secured to the support flange via: one or more fasteners, and a retaining member at the distal end of the spoke, the retaining member defining an abutment face facing an axial direction relative to the central axis and circumferentially overlapping the support flange, a portion of the support flange located axially between the distal end of the spoke and the abutment face.
2. The aircraft engine of claim 1, wherein the retaining member is a hook at the distal end of the spoke, the hook having a first hook section protruding axially from the distal end and a second hook section protruding transversally from the first hook section, the abutment face defined by the second hook section.
3. The aircraft engine of claim 2, wherein the hook and the spoke are parts of a single monolithic body of the spoke.
4. The aircraft engine of claim 1, wherein the support flange defines a member-receiving aperture sized to receive the retaining member.
5. The aircraft engine of claim 1, wherein the retaining member defines a slot for receiving the portion of the support flange, the slot facing a circumferential direction opposite to a direction of rotation of the turbine rotor.
6. The aircraft engine of claim 1, wherein the scroll case includes vanes extending in a direction having an axial component relative to the central axis and across the conduit.
7. The aircraft engine of claim 6, wherein each of the spokes extends within a respective one of the vanes.
8. The aircraft engine of claim 7, wherein each of the spokes are free of connection to the

respective one of the vanes.

9. The aircraft engine of claim 1, wherein the turbine support case includes a wall extending around the central axis, each of the spokes protruding from the wall.
 10. The aircraft engine of claim 9, wherein the wall axially overlaps at least a portion of the turbine, the turbine support case having a rear flange secured to a flange of the exhaust case.
 11. A turbine assembly, comprising: a turbine including a turbine rotor rotatable about a central axis; a support structure; a scroll case for receiving combustion gases and for directing the combustion gases to the turbine, the scroll case having a conduit extending around the central axis; and a turbine support case having spokes distributed around the central axis, each of the spokes extending through the conduit of the scroll case and radially supported by the support structure via: one or more fasteners, and an axial locking engagement defined between distal ends of each of the spokes and a portion of the support structure.
 12. The turbine assembly of claim 11, wherein the axial locking engagement is defined by hooks provided at distal ends of each of the spokes.
 13. The turbine assembly of claim 12, wherein the hooks have first hook sections protruding axially from the distal ends and second hook sections protruding transversally from the first hook sections, the second hook sections axially overlapping the portion of the support structure.
 14. The turbine assembly of claim 12, wherein the hooks and the spokes are parts of a single monolithic body of the turbine support case.
 15. The turbine assembly of claim 12, wherein the portion of the support structure is an annular flange defining apertures, the hooks received through the apertures.
 16. The turbine assembly of claim 15, wherein the hooks define slots for receiving a portion of the annular flange.
 17. The turbine assembly of claim 16, wherein the slots are facing a circumferential direction opposite to a direction of rotation of the turbine rotor.
 18. The turbine assembly of claim 11, wherein the scroll case includes vanes extending in a direction having an axial component relative to the central axis and across the conduit.
 19. The turbine assembly of claim 18, wherein each of the spokes extends within a respective one of the vanes.
 20. The turbine assembly of claim 19, wherein each of the spokes are free of connection to the respective one of the vanes.
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